

→ Wrapper Classes in Java :-
 Java has **primitive data type** such as :-
 byte
 short
 int
 long
 float
 double
 char
 boolean

• A Wrapper Class is a class that provides an **object representation** of a **primitive data type**.

In simple words :-

Wrapper classes **convert** **primitive data** into **objects**.

Eg;

int a = 10; // primitive

Integer b = 10; // wrapper object

here ;

int → Integer
 char → Character
 boolean → Boolean

→ Primitive Type and Wrapper Classes :
 This is the most important table to remember :

Primitive Data type Wrapper Class

byte	Byte
short	Short
int	Integer
long	Long
float	Float
double	Double
char	Character
boolean	Boolean

→ Why do we need wrapper classes?

Java collections works with objects, not primitive data types.

Eg ;

We cannot write :

```
ArrayList<int> list ; // error
```

```
ArrayList<Integer> list ; // correct
```

eg;

```
import java.util.Scanner;
```

```
class pogo {
```

```
    psum {
```

```
        ArrayList<Integer> list = new ArrayList<>();
```

```
        list.add(10);
```

```
        list.add(20);
```

```
        list.add(30);
```

```
        Sout(list);
```

```
    }
```

```
}
```

output:

[10, 20, 30]

Here integer is the wrapper class of int.

→ Primitive vs Wrapper Object:

Primitive:

```
int x = 10;
```

here, x directly stores the primitive value.

Wrapper :

Integer y = 10;

here y refers to an Integer object containing the value 10.

→ Important Wrapper Classes in java

(A) Integer :
 Wrapper class of int.

Integer x = 100;

Useful method :

Integer.parseInt("100");

Eg;

```
String s = "123";
int n = Integer.parseInt(s);
System.out.println(n+10);
```

output :

133

parseInt() converts a String into an int.

(B) Double:

Wrapper class for double.

Double x = 10.5;

Eg;

String s = "10.5";

double d = Double.parseDouble(s);
Sout(d);

(C) Character:

Wrapper class for Char.

Character ch = 'A';

Useful method:

Character.isLetter(ch)

Character.isDigit(ch)

Character.isUpperCase(ch)

Character.isLowerCase(ch)

Character.toUpperCase(ch)

Character.toLowerCase(ch)

Eg;

char ch = 'A';

Sout(Character.isLetter(ch));

Sout(Character.isUpperCase(ch));

(D) Boolean:

Wrapper class for boolean

Boolean result = true;

Boolean x = true;

out(x);

(E) Byte, Short and Long:

Byte b = 10;

Short s = 100;

Long L = 100000L;

(F) Float:

Wrapper class for float.

Float f = 10.5f;

~~for more classes~~

